

UGTS LIBRARY EDITION

VOLUME 2

SHAPES, PATHS, FIELDS AND MOTION

How a rule-world describes form and movement

Concrete play, diagrams, error bands and adult formal bridges

PATTERN

A rule that can make a family of shapes.

PATH

A route with a parameter and direction.

FIELD

A value returned at each query point.

FRAME

A local “forward, side and up” guide.

GUARD BAND

A typed boundary plus error or uncertainty.

PACKING

Finite keys and labels, never the whole world.

DESCRIBE BEFORE YOU CHANGE

Geometry answers queries; the event chain decides authority

BIG IDEA

UGTS can describe shapes, routes, fields and motion without turning a drawing or rendered frame into the authoritative world. Description supports a proposal; the six-question chain still decides change.

SHAPE

What points belong to a form, boundary or support?

PATH

Where does a parameter place an object, and how does direction change?

FIELD

What value, sign, gradient or status appears at a query point?

MOTION

What is the state at a requested time, with a declared model and error contract?

ACCESS ROUTE

Trace, build, listen, point or describe. A reader may use string, blocks, raised lines, sound cues, movement or a screen. The mathematical role remains the same.

PATTERN, PATH, FIELD OR PICTURE?

Four related things that must not be silently merged

THING	FORMAL ROLE	EXAMPLE
Pattern	A versioned rule with parameters and a domain.	A rose family, spiral law, spline or superellipse.
Path	A parameterized route through positions and possibly orientations.	A curve used for motion, calibration or a sweep.
Field	A function that returns a value at each query point.	Inside/outside sign, temperature, confidence or arrival time.
Picture	A downstream projection for people or displays.	SVG, Canvas, Bayer pixels, camera preview or report figure.

WHY THIS MATTERS

The same curve may be used as a boundary, trajectory, support or calibration sweep. The role must be declared. A picture of the curve is still only a presentation.

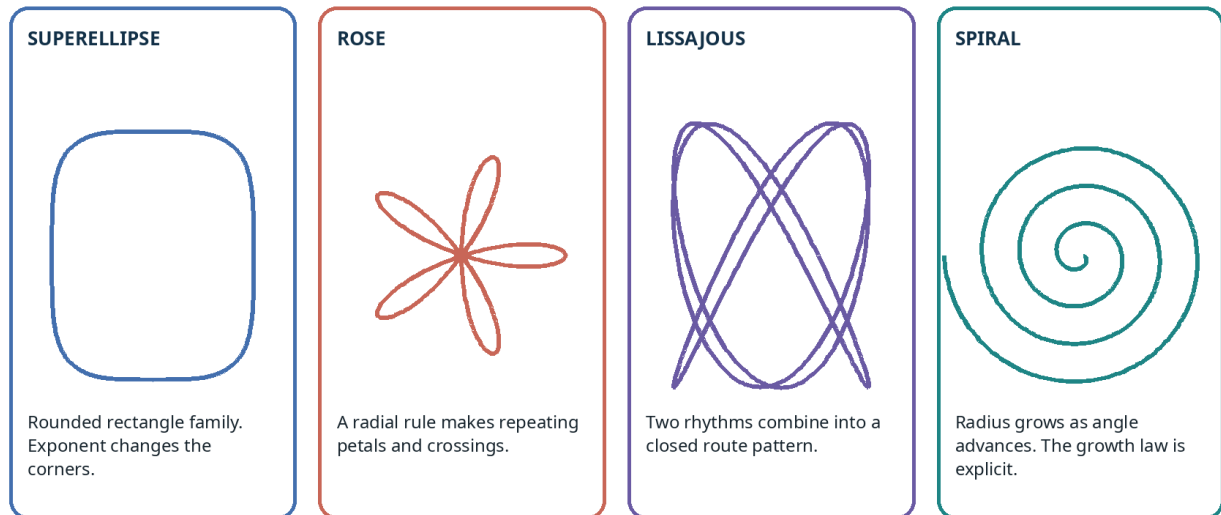
Source note: Source basis: UGTS-KC 2.0 pattern instance contract and query set; KC Two Hands 3.0 and Bayer Direct keep presentation downstream.

PATTERN FAMILIES

One rule, many parameter choices

PATTERNS ARE RULES FOR MAKING SHAPES

A pattern is not just a picture. It has parameters, a domain, a frame and limits.



Formal bridge: UGTS-KC 2.0 stores a pattern as a versioned operator with parameters, frame, support, tags and a validation contract.

PLAY IDEA

Use a bendable wire or string. Change one parameter at a time: corner roundness, petal count, rhythm pair or radial growth. Name what changed and what stayed invariant.

HONEST LIMIT

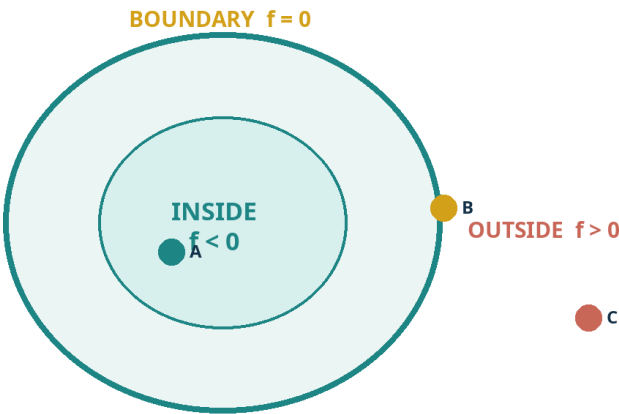
A famous mathematical name does not prove a material property, biological law, self-assembly effect or product performance. Those require separate models and measurement.

FIELDS: INSIDE, BOUNDARY AND OUTSIDE

A number or status can live at every query point

A FIELD GIVES A VALUE AT EVERY QUERY POINT

The sign may classify inside, boundary and outside. "Distance" is an extra capability, not an automatic label.



A zero is only an event candidate. Support, compatibility, guard policy, error and lineage still decide whether anything changes.

GENERIC IMPLICIT FIELD

May support sign, threshold, gradient or guard queries. It is not automatically exact Euclidean distance.

EXACT SIGNED DISTANCE

Adds a stronger contract: magnitude equals shortest distance under the declared primitive and transform conditions.

EVENT RULE

The zero boundary is a candidate. A change still requires support, compatibility, accepted guard status, confidence/error and deterministic commit.

SUPPORT SHAPES

Ask a small local question before a large exact one

BOX / AABB Fast finite bounds for scene or map candidates. A box is not the final semantic answer.	SPHERE Local radial support or exact sphere relation when center and radius are declared.
CONE Radial reach plus angular direction. Useful for views, sensors and directional relevance.	SHELL An ordered radial interval for multiscale relevance or lookup.

Hybrid candidate path

ORDER conventional bounds / index -> local support -> compatibility -> exact pattern, field or trajectory query -> guard classification -> proposal
KILL CRITERION Use a conventional index instead when the combined support/compatibility system is slower, larger or less accurate at equal error.

Source note: Sources: UGTS-GN 1.1 local radial-angular support; UGTS-KC 2.0 support and kill criteria; Two Hands 3.0 AABB/BVH hybrid path.

MOTION: WHERE, HOW FAST, AND HOW THAT CHANGES

Position alone is not a motion model

CHILD ROUTE

A moving object can have a place, a speed, a turning direction and a “change of change.” UGTS stores only the fields needed by the selected motion profile and keeps the model explicit.

STATE PART	PLAIN MEANING
position	Where the state is at a time.
velocity	How position changes.
acceleration	How velocity changes.
curvature	How strongly a path turns per distance.
torsion	How a three-dimensional path twists out of a plane.
mode	Which continuous or hybrid rule is active.

FORMAL EXAMPLE

For constant acceleration: $x(t) = p_0 + v_0 \cdot dt + \frac{1}{2} \cdot a \cdot dt^2$ and $v(t) = v_0 + a \cdot dt$. A relation $f(x)=0$ becomes a time guard $g(t)=f(x(t))$.

BOUNDARY

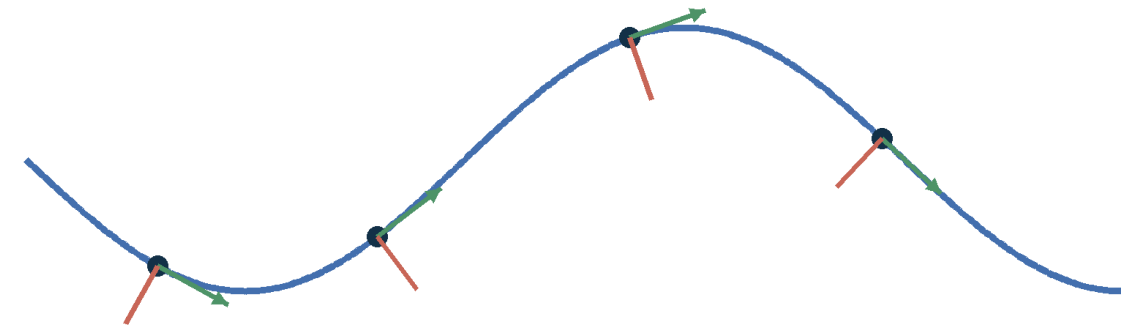
A coordinate chart may change component formulas. A physical force or torque requires a declared mechanics model; coordinate vocabulary does not create dynamics.

MOVING FRAMES

Carry a stable local direction along a route

A MOVING FRAME CARRIES “FORWARD” WITH THE PATH

Position says where. A tangent says forward. A normal gives a sideways reference.



green = tangent / forward

coral = normal / side

Frame status matters near stops, cusps and very low curvature.

FRENET FRAME

Uses derivatives and curvature. It can become unstable where speed or curvature approaches zero.

BISHOP / PARALLEL TRANSPORT

Carries a rotation-minimizing frame from an initial direction. It requires a numerical transport tolerance.

WHY THE STATUS MATTERS

A frame query should return both orientation and status. “Undefined,” “low curvature,” or “fallback used” is more truthful than inventing a direction.

LOOPS, CROSSINGS AND WINDING

The same coordinate can belong to a different trip

Imagine walking around a circular track three times. You return to the same painted point, but your winding count and route history are different. A projected crossing may also show two route branches that are not actually connected.

POSITION The current coordinate.	PHASE Where you are inside one repeating cycle.
WINDING How many whole cycles have been completed.	ROUTE CLASS Which obstacle, sheet, braid or portal history distinguishes the path.
COMPATIBILITY RULE Same position plus different sheet, phase, mode or route class may mean no coupling. A visible self-intersection is not automatically a collision.	

Source note: Sources: UGTS-KC 2.0 winding, covering and homotopy descriptors; SCLP 3.6.2 phase plus winding and explicit topology profiles.

GUARD BANDS AND UNCERTAINTY

Sometimes the truthful answer is “not classified yet”

Suppose a doorway is measured as 79.8 cm with a 0.5 cm uncertainty, while a rule requires at least 80.0 cm. The interval overlaps the threshold. A careful system should not silently call it open or blocked.

STATUS	INTERVAL TEST	ACTION
clearly below	upper bound < threshold	Classify below.
clearly above	lower bound > threshold	Classify above.
overlaps guard	interval touches the threshold band	Return ambiguous; remeasure or follow policy.
too uncertain	width exceeds event margin	Reject or simplify the claim.

ONE-BIT JITTER LESSON

A deterministic bit may choose a tiny signed perturbation or route. It is admissible only when its whole interval stays inside the event margin and cannot change the event class or order.

Source note: Source basis: SCLP 3.6.2 certified jitter intervals and sweep bounds; KC Atlas and 4.0 spatial ledger use conservative observation intervals.

SWEEPS AND FINITE SAMPLING

Move a shape along a route without pretending every sweep has a closed form

CHILD ROUTE

Move a cardboard cone or circle along a string. The swept region is every place touched during the trip. If we check only a finite set of stops, we also need a bound on what could happen between them.

EXACT PRIMITIVE

A declared sphere or finite cone may have an exact local relation.

CERTIFIED SWEEP

Sample minima plus a Lipschitz or motion bound produce an interval for the true sweep relation.

DO NOT OVERCLAIM

A translational sweep certificate is not a universal exact signed-distance formula for arbitrary rotation, deformation or grammar-generated motion. Extra motion terms or a specialized solver may be required.

FINITE PACKING AND KEYS

A small address can name a cell without storing the whole world

SCLP supplies two explicit 64-bit layouts for quantized log-radius, angle, modular time and hinge angle. One uses contiguous fields; one interleaves bits in a Morton-style schedule. Both round-trip the same quantized tuple, but the words are intentionally different.

QUANTIZE Map a declared continuous range to finite integer cells and record the error.	PACK Place integer fields into a declared fixed-width layout.
PREFIX Each appended radix bit refines one scheduled interval; it does not directly halve the physical coordinate.	DECODE Recover the quantized tuple and its declared range/error, not imaginary infinite detail.
COMPRESSION BOUNDARY A width ratio is not equal-information compression unless the records preserve the required semantics, reconstruction and error. One sign bit does not retain distance magnitude or uncertainty; a seed does not recreate unstored photons.	

Source note: Sources: UGTS-GN 1.1 packed precision and compression contracts; SCLP 3.6.2 key layouts and pointerless-storage audit; 4.1/4.1.1 seed-plus-evidence boundary.

APPLICATION: A DETERMINISTIC GAME WORLD

Geometry becomes playable without becoming the sole authority

PHASE	ROLE
input	Resolve named actions into a serializable frame.
controller	Turn actions into intended movement or dash.
physics	Integrate bodies under an explicit fixed step.
bounds	Clamp, bounce or wrap under a declared rule.
collision	Compute candidate pairs and manifolds with filters.
gameplay	Apply hazards, collectibles and win/lose rules through events.
late / camera / flush	Update lifetime, follow camera and perform deferred destruction.

PICTURE VERSUS WORLD

Vector art, particles, HUD, sound and Canvas output are downstream feedback. Canonical snapshots, events, state hashes and project records carry the testable game identity.

Source note: Source: KC Elizabeth 3.9 deterministic GameWorld, vector art, input, animation, collision, saves and self-contained HTML5 runtime.

THREE HANDS-ON ACTIVITIES

Use household materials and preserve reader control

ACTIVITY	STEPS / CONCEPT
STRING PATH	Tape a string route. Move an arrow token along it. At three points, place a “forward” arrow and a “side” arrow. Ask what happens at a sharp turn or stop.
FIELD FLOOR	Mark INSIDE, BOUNDARY and OUTSIDE regions with mats or sounds. Place query tokens and let the reader classify, skip or ask for help.
PACKING BOX	Use four compartments with fixed capacities. Place coded cards, decode them and compare a contiguous order with an interleaved order. Keep a separate card for uncertainty.

ACCESS OPTIONS

Raised string, textured mats, spatial audio, object cues, switch scanning, partner-assisted placement and recorded descriptions can all express the same concepts.

PHYSICAL BOUNDARY

Choose safe material sizes and attachment methods. These are informal learning activities, not manufacturing drawings or certified toys.

FORMAL BRIDGE

Compact dictionary for builders and adult learners

TERM	CONTRACT
pattern	Versioned operator + parameters + frame + support/tags + validation contract.
parametric curve	$\gamma: U \rightarrow \mathbb{R}^n$ with declared parameter interval and derivatives/status.
implicit field	$f: \mathbb{R}^n \rightarrow \mathbb{R}$ with declared sign, threshold and exactness capability.
state_at(t)	Closed expression or bounded integration with a model and error/work contract.
frame_at	Frenet, Bishop or body frame plus validity/fallback status.
route_class	Winding, homotopy, braid, cell or chart descriptor that supplements coordinates.
guard interval	Residual/error interval classified against an event margin.
packed key	Invertible finite quantized address under a declared layout and range.

4.2 CONNECTION

The general OperatorSpec can record each of these operators' domain, codomain, units, shape, exactness, effects, dependencies and order. A mature domain runtime may remain the implementation while 4.2 supplies schema and handoff discipline.

SOURCE REGISTER AND HONEST LIMITS

What this volume can and cannot establish

SOURCE	USED FOR
UGTS-KC 2.0	patterns, fields, topology, SE(2)/SE(3), frames, timing, dynamics and kill criteria
UGTS-KC 3.6	implicit/parametric definitions, hinges and explicit operation order
SCLP 3.6.2	finite cone/sphere relations, sweep intervals, log-polar calculus, winding and 64-bit keys
KC Two Hands 3.0	spatial indexing, geometry compilation, independent error budgets and downstream rendering
KC Elizabeth 3.9	vector/game runtime, fixed-step phases and persistence
KC Atlas / 4.0 / 4.1	uncertainty intervals, observation/authority separation and seed-plus-evidence boundaries
UGTS-KC 4.2	general operator and order records

NO MAGIC GEOMETRY

Topology labels do not replace mechanics, conservation laws, sensor calibration or evidence. A finite key does not store infinite detail. A picture does not become authoritative state.

EVIDENCE RULE

Tests and demonstrations establish internal properties of their declared implementations. Application accuracy, physical performance and safety require target-specific measurement and review.

VOLUME 2 COMPLETE

One consecutive library volume, not the end of the project

What this volume establishes

- Pattern, path, field and picture are separated by role.
- Inside/boundary/outside queries are taught without calling every field exact distance.
- Motion, moving frames, winding and route identity are connected to concrete examples.
- Guard intervals teach that “ambiguous” can be the truthful result.
- Finite keys and seed records are presented with reconstruction and evidence boundaries.

Next volume in the sequence

VOLUME 3 - IDENTITY, RULES, MEMORY AND REPLAY

The next book explains why coordinates and pictures are not identity, how definitions live inside the substrate, and how proposals, commits, hashes, checkpoints and spatial evidence form a trustworthy remembered world.

EDITION CONTROL

UGTS Library Edition, Volume 2, edition 0.1. This educational layer explains source releases and does not modify their formal version numbers.